

Advanced (Carolina Reaper)

- Q1.** The graph of  $y = f(x)$  is shifted 2 units to the right. The new equation is  
(A)  $y = f(x - 2)$   
(B)  $y = f(x + 2)$   
(C)  $y = f(x) + 2$   
(D)  $y = f(x) - 2$   
(E)  $y = f(x - 1)$
- Q2.** If  $y = f(x)$  is reflected over the x-axis, the new equation is  
(A)  $y = f(x) + 1$   
(B)  $y = f(x) - 1$   
(C)  $y = -f(x)$   
(D)  $y = f(-x)$   
(E)  $y = f(x + 1)$
- Q3.** The graph of  $y = f(x)$  is compressed vertically by a factor of 2. The new equation is  
(A)  $y = 2f(x)$   
(B)  $y = f(2x)$   
(C)  $y = \frac{1}{2}f(x)$   
(D)  $y = f(x) + 2$   
(E)  $y = f(x) - 2$
- Q4.** If  $y = f(x)$  is shifted 3 units up, the new equation is  
(A)  $y = f(x) + 3$   
(B)  $y = f(x) - 3$   
(C)  $y = f(x + 3)$   
(D)  $y = f(x - 3)$   
(E)  $y = f(x) - 1$
- Q5.** The graph of  $y = f(x)$  is reflected over the y-axis. The new equation is  
(A)  $y = f(-x)$   
(B)  $y = -f(x)$   
(C)  $y = f(x) + 1$   
(D)  $y = f(x) - 1$   
(E)  $y = f(x + 1)$
- Q6.** If  $y = f(x)$  is stretched horizontally by a factor of 3, the new equation is  
(A)  $y = f(\frac{1}{3}x)$   
(B)  $y = f(3x)$   
(C)  $y = 3f(x)$   
(D)  $y = f(x) + 3$   
(E)  $y = f(x) - 3$
- Q7.** The graph of  $y = f(x)$  is compressed horizontally by a factor of 2. The new equation is  
(A)  $y = f(2x)$   
(B)  $y = f(\frac{1}{2}x)$   
(C)  $y = 2f(x)$   
(D)  $y = f(x) + 2$   
(E)  $y = f(x) - 2$
- Q8.** If  $y = f(x)$  is shifted 2 units to the left and 1 unit up, the new equation is  
(A)  $y = f(x + 2) + 1$   
(B)  $y = f(x - 2) + 1$   
(C)  $y = f(x) + 2$   
(D)  $y = f(x) - 2$   
(E)  $y = f(x + 1) - 2$

Open Ended Questions (FRQ)

- Q9.** Graph the function  $y = f(x) = x^2$  and its transformation  $y = f(x - 2) + 1$ . Compare the two graphs and describe the transformations that occurred.
- Q10.** Given the function  $y = f(x) = |x|$ , describe the transformations that would result in the graph of  $y = -f(2x) + 3$ . Show the graph of the original function and its transformation.

Chef's Tips

- Tip 1: Review the parent function before applying any transformations
- Tip 2: Use graph paper to help visualize the transformations
- Tip 3: Check your work by graphing the original and transformed functions